INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

CEREAL BARS, (OPERATIONAL RATION COMPONENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cereal bars for use by the Department of Defense as a component of operational rations.

1.2 Classification. The cereal bars shall be of the following types and styles as specified (see 6.1).

Type I - Shredded wheat cereal bar
Type II - Corn flakes cereal bar
Type III - Wheat flakes cereal bar
Type IV - Malted wheat granules cereal bar
Type V - Bran flakes cereal bar
Type VI - Oat cereal biscuit bar

Style A - 1.0 by 3.0 by 0.6-inch bar weighing 32 to 38 grams
Style B - 1.0 by 3.0 by 1.0-inch bar weighing 43 to 53 grams

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
A-A-20000	-	Cereal, Prepared; Ready-to-Eat
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular, Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

MILITARY

MIL-F-43231	-	Food Packet, Survival, General Purpose, Packaging And Assembly Of

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Instant Nonfat Dry Milk (7 CFR Part 58)

(Copies are available from the Dairy Standardization Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Copies are available from the Chief, Standardization Branch, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 3944, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

U.S. Standards of Identity for Dried Eggs (21 CFR Part 160)

U.S. Standards of Identity for Margarine (21 CFR Part 166)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastic by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Cereals. Cereals except shredded wheat, shall be fortified in accordance with the Federal Food, Drug, and Cosmetic Act. At time of use the cereal shall contain no more than 3 percent moisture. The cereals shall comply with the following types, classes and styles of A-A-20000:

a. Shredded wheat biscuits, type I, class 1, style A, B, C, or D.
b. Wheat flakes, type I, class 3.
c. Bran flakes, type I, class 5.

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d. Malted wheat cereal granules, type I, class 10 (see 6.3.1).
e. Corn flakes, type II, class 1.
f. Oat cereal biscuits, type IV, class 3, style A (see 6.3.2).

3.2.2 Cream substitute, non-dairy, dry. The dry, non-dairy cream substitute shall be a white to light cream color, free flowing powder that possesses the capability of rapid and complete solubility. The cream substitute shall contain corn syrup solids, partially hydrogenated vegetable oils, sodium caseinate, mono and di-glycerides, dipotassium phosphate, artificial flavors and colors. The cream substitute shall contain not more than 4.9 percent moisture, not less than 31 percent fat, and not less than 95 percent shall pass through a U.S. Standard No. 10 sieve. Each product lot shall be accompanied by a certificate stating that the product is negative for the presence of Salmonella as tested by a USDA-AMS recognized laboratory.

3.2.3 Eggs, whole, dried. The dried whole eggs shall be processed and labeled in accordance with the USDA Regulations Governing the Inspection of Eggs and Egg Products. The dried whole eggs shall be USDA certified salmonella free and shall conform to the FDA Standards of Identity for Dried Eggs. The dried whole eggs shall be of the latest date of pack. The USDA certificate shall accompany egg products shipped between plants.

3.2.4 Flour, wheat. Wheat flour shall be free-flowing, enriched, general purpose white (bleached) flour.

3.2.5 Glycerin. Glycerin shall comply with the requirements of the Food Chemicals Codex.

3.2.6 Lecithin, liquid. Liquid lecithin shall be a bland, light amber, clear, refined liquid with a low viscosity (1500 maximum centipoise at 25°C) and shall comply with the requirements of the Food Chemicals Codex (see 6.3.3).

3.2.7 Margarine. Margarine shall comply with the FDA Standard of Identity for Margarine. The margarine shall be manufactured from corn oil and shall possess a fine and pleasing flavor. The margarine shall be salted. The body shall be smooth, firm and homogeneous. The product shall possess a uniform medium yellow color.

3.2.8 Milk, dry, nonfat, instant. The instant nonfat dry milk shall be U.S. Extra Grade according to the U.S. Standards for Instant Nonfat Dry Milk.

3.2.9 Salt. Salt shall be white, refined sodium chloride with or without anti-caking agents and shall comply with the requirements of sodium chloride of the Food Chemicals Codex.

3.2.10 Silica. Silica shall comply with the requirements for Silicon Dioxide of the Food Chemicals Codex.

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3.2.11 Sodium bicarbonate. Sodium bicarbonate shall comply with the requirements of the Food Chemicals Codex.

3.2.12 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof.

3.2.13 Titanium dioxide. Titanium dioxide shall be water dispersible and shall comply with the requirements of the Food Chemicals Codex.

3.2.14 Water. Water used for washing, formulation and processing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Formulation and preparation of cereal matrix. The cereal matrix shall be formulated and prepared as follows:

Ingredient	Percent by weight
Flour, wheat	33.60
Cereal (see type specified in 6.1)	23.00
Margarine	21.50
Sugar, white, granulated	11.20
Water	8.00
Eggs, whole, dried, free-flowing	1.60
Lecithin, liquid	0.50
Sodium bicarbonate	0.35
Salt	0.25

NOTE: The following matrix preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time, temperature and moisture requirements specified for the prepared product are still applicable).

a. Mix dried eggs with water. Let stand 1/2 hour.

b. Cream margarine and lecithin together.

c. Add sugar and rehydrated eggs to the margarine/lecithin mixture and mix well.

d. Add dry ingredients (except cereal) and combine well in a Hobart type mixer.

e. Add cereal and mix until just blended.

f. Chill the dough 1 to 2 hours or overnight.

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g. Bake the matrix at 250°F until moisture content is less than 2 percent (see note). With a thickness of 1/2 inch, baking should take 10 to 15 minutes. The baked matrix shall be light to medium tan in color with no areas that are brown, scorched, or burnt. When using corn flakes, the color may be a golden tan to tan.

NOTE: Silica or flour may be used on cooking surfaces if needed.

3.3.1.1 Handling of cooked cereal matrix. When cooled, store cooked matrix in vacuumed sealed container until ready to use. Matrix may be held under vacuum in accordance with good commercial practice for up to 30 days before forming into bars. If matrix is further processed into bars within 48 hours of cooking, vacuum sealing is not required. Product shall be held in accordance with good commercial practice at room temperature such that matrix moisture level does not exceed 2 percent.

3.3.2 Formulation, preparation, and compression of cereal bars. The cereal bars shall be formulated, prepared, and compressed as follows:

Ingredient	Percent by weight
Cereal matrix (see type specified in 6.1)	88.00
Cream substitute, non-dairy, dry	4.90
Milk, dry, nonfat, instant	4.90
Glycerin	2.00
Titanium dioxide	0.20

a. Mix cereal matrix, cream substitute, nonfat dry milk, and titanium dioxide in a suitable mixer until matrix is reduced to about 1/2 inch pieces. (Matrix may be ground through a screen not smaller than 1/2 inch prior to mixing).

b. Gradually add glycerin and mix until just blended.

c. The cereal bar mixture shall be compressed to produce bars complying with finished product requirements (see 3.6).

3.3.3 Pouch filling and sealing.

3.3.3.1 Food Packet, Survival, General Purpose. For the Food Packet, Survival, General Purpose, cereal bars shall be filled and sealed in pouches to comply with finished product requirements and to the requirements specified in MIL-F-43231.

3.3.3.2 Food Packet, Long Range Patrol, (Improved). For the Food Packet, Long Range Patrol (LRP), cereal bars shall be filled and sealed in pouches to comply with finished product requirements and to the requirements specified in 5.1.1.

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3.4 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, stale, sour, rancid, musty or moldy.

c. There shall be no color foreign to the product.

d. There shall be no compression streaks or stains.

e. The dimensions and weights shall be as follows:

Style A - Dimensions: 1.0 (± 0.1) inch by 3.0 (± 0.1) inches by 0.6 (± 0.1) inch thick. Weight: 35 ± 3 grams.

Style B - Dimensions: 1.0 (± 0.2) inch by 3.0 (± 0.2) inches by 1.0 (± 0.2) inch thick. Weight: 48 ± 5 grams.

f. The color shall be light tan to medium tan, with corn flakes a golden tan.

g. The odor and flavor shall be slightly sweet, slight dairy or creamer, and slight cereal typical of the type of cereal used.

h. The bar shall be crumbled by hand into pieces equal to or less than 1/2 inch diameter and mixed into 50 mL of water and rehydrated to a milky or cream-colored liquid with small pieces of cereal within 15 to 45 seconds.

i. No pouch shall contain a bar which has been broken or crumbled (more than three pieces).

j. The moisture content shall be not greater than 5.0 percent.

3.4.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatibility and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.1.2 Pouch material certification. The material listed below may be accepted on the basis of a contractor's certificate of conformance to the indicated requirements. In addition, compliance to the requirements for inside pouch dimensions and dimensions of manufacturer's seals may be verified by certificate of conformance.

Requirement	Requirement paragraph	Test procedure
Thickness of films for laminated material	5.1.1.1.1 and 5.1.1.2.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	5.1.1.1.1 and 5.1.1.2.1	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.1.1 and 5.1.1.2.1	Laboratory evaluation.
Color of laminated material	5.1.1.1.1 and 5.1.1.2.1	Visual examination

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 32 grams or greater than 38 grams for the General Purpose Survival Food Packet or a net weight of less than 43 grams or greater than 53 grams for the LRP shall be classified as a minor defect. The results shall be reported to the nearest gram. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

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TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, cut, hole, or abrasion through one or more layers in the pouch material or leakage through any heat seal 2/
102		Foldover wrinkle extending into the seal such that the effective closure seal is reduced to less than 1/16 inch 3/
103		Presence of entrapped product that reduces the effective closure seal to less than 1/16 inch 3/
104		Presence of delamination 4/
105		Unclean pouch 5/
106		Labeling is missing, incorrect, or illegible
107		Any impression or design on the heat seal surfaces which conceals or impairs visual detection of seal defects 6/
108		Less than 3/16 inch between inside edge of tear notch and inside edge of seal
109		Closure seal width not as specified
110		Closure seal not located as specified
111		Presence of stress cracks in the aluminum foil 7/ 8/
	201	Tear notch or serrations missing or not located as specified
	202	Depth of tear notch or serrations not as specified
	203	Presence of delamination 4/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Pinholes or breaks inherent to the manufacturing process of the aluminum foil shall not be scored as a defect.

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3/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4/ Delamination defect classification:

Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise-counterclockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the laminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as minor defects.

5/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the pouch with a clean dry cloth.

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b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

6/ If doubt exists as to whether or not the sealing equipment leaves an impression or design on the closure seal surface that could conceal or impair visual detection of seal defects, samples shall be furnished to the contracting officer for a determination as to acceptability.

7/ Applicable to form-fill-seal pouches only.

8/ To test for stress cracks, the inside surface of both tray-shaped bodies shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the pouch material in the form of a curved or straight line greater than 2 mm in length shall be evidence of the presence of stress cracks in the aluminum foil.

4.5.5 Pouch vacuum examination. The filled and vacuum sealed pouches shall be visually examined for conformance to the vacuum requirement in 5.1.1.1.3 and 5.1.1.2.2 not less than 96 hours after filling and sealing. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, while simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of compression streaks or stains
102		Dimensions of bar(s) not as specified
103		Color of bar(s) not as specified

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TABLE II. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
104		Odor and flavor of bar(s) not as specified
105		Bar(s) does not rehydrate as specified (see 3.6h)
	201	Broken or crumbled bars (more than three pieces) 3/

1/ The presence of any foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot (see 3.4.1).

3/ A broken or crumbled bar is a bar which either:

a. has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume, or

b. has more than 1/8th of the volume of the bar crushed.

4.5.7 Internal pressure test. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size and individually tested for internal pressure resistance. The internal pressure resistance shall be determined by pressurizing the pouches while they are restrained between two rigid plates spaced 1/2 ± 1/16 inch apart. If a three-seal tester (one that pressurizes the pouch through an open end) is used, the closure seal shall be cut off for testing the side and bottom seals of the pouch; for testing of the closure seal, the bottom seal shall be cut off. The pouches shall be emptied prior to testing. If a four-seal tester (designed to pressurize filled pouches by use of a hypodermic needle through the pouch wall) is used, all four seals can be tested simultaneously. The distance between rigid restraining plates on the four-seal tester shall be equal to the thickness of the product + 1/16 inch. Pressure shall be applied at the approximate uniform rate of 1 pound per square inch gauge (psig) per second until 14 psig pressure is reached. The 14 psig pressure shall be held constant for 30 seconds and then released. The pouches shall then be examined for separation or yield of the heat seals. Any rupture of the pouch or evidence of seal separation greater than 1/16 inch in the pouch manufacturers seal shall be considered a test failure. Any seal separation

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that reduces the effective closure seal width to less than 1/16 inch (see table I) shall be considered a test failure. The lot size shall be expressed in pouches. The sample unit shall be one pouch. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Moisture content testing. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size. The contents of each pouch shall be individually blended to uniformity using a blender or a food processor. The blending must be rapid and conducted in such a way that minimum heat is transferred to the product and that the product has minimum exposure to atmospheric moisture. The blended samples shall be tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 925.45A except that the temperature-time cycle shall be modified by using a temperature of 70°C for 16 hours at a pressure of not more than 100 mm mercury. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: Marking missing, incorrect, or illegible
Cushioning material missing or not as specified

Minor: More than 40 pounds packed in shipping container
Pouches not packed flat in layers

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be unit packed in pouches in accordance with 5.1.1.1 or 5.1.1.2 for LRP applications or in accordance with MIL-F-43231 for Food Packet, Survival, General Purpose applications.

5.1.1.1 Preformed pouch.

5.1.1.1.1 Pouch material. The performed pouch shall be a 3-ply laminate consisting of, from inside to outside, 0.003 inch thick polyethylene, 0.0007 inch thick aluminum foil, and 0.0005 inch thick polyester. All tolerances for thickness of pouch materials shall be plus or minus 20 percent. Color shall be applied completely and uniformly to either the exterior or interior surfaces of the polyester lamina. The pouches shall be colored overall with a color in the range of green 34079 through 34087, or 24052 through 24087, or brown 30045 through 30118 (excluding 30109) of FED-STD-595. The color application process shall not adversely affect the material performance. The pouch material shall not impart any undesirable odor or flavor to the product

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and shall be FDA approved for food use. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.1.1.2 Pouch construction. The pouch shall be a flat style preformed pouch having inside dimensions of 2-3/4 inches by 5-3/4 inches (± 1/8 inch in each dimension). The first dimension shall be the opening of the pouch between the heat sealed sides. The pouch shall be made by heat sealing three edges with 3/8 inch (+ 3/16 inch, - 1/8 inch) wide seals. A "V", "C", or block "U" tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 5/16 inch to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches provided that the serrations are sharp (no plastic tailings exist) and the serration depth and the minimum seal width at the serrated edges are in accordance with the above requirements. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8 inch).

5.1.1.1.3 Pouch filling and sealing. One bar of style B shall be placed in the pouch in a manner as to avoid puncturing the pouch material or contaminating the heat seal area. The filled pouches shall be sealed under a vacuum of not less than 23 inches of mercury. The sealed pouch shall show no loss of vacuum when examined as specified in 4.5.5. The filled pouches shall have a 3/8 inch (± 1/8 inch) heat seal. If thermal impulse closure seals are used, a seal width of 1/8 to 7/16 inch will be acceptable. When tested as specified in 4.5.7, the filled and sealed pouches shall withstand an internal pressure of 14 psig for 30 seconds without rupture or seal separation greater than 1/16 inch or seal separation that reduces the effective closure seal width to less than 1/16 inch. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of fold over wrinkles that reduce the effective closure seal to less than 1/16 inch.

5.1.1.2 Form-fill-seal pouch.

5.1.1.2.1 Pouch material. The form-fill-seal pouch shall consist of a formed tray-shaped body with a tray-shaped heat sealable cover. The tray-shaped body and the tray-shaped cover shall be fabricated from 3-ply flexible laminate barrier material consisting of, from outside to inside, 0.0009 inch thick oriented polypropylene bonded to 0.0007 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene bonded to 0.003 inch thick ionomer or a blend of not less than 50 percent linear, low density polyethylene and polyethylene. All tolerances for thickness of pouch materials shall be plus or minus 20 percent. The linear low density

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polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cc in accordance with ASTM D 1505. Alternatively, 0.0005 inch thick polyester may be used in place of the oriented polypropylene as the outer ply of the laminate. The color requirements of the exterior of the laminates shall be as specified in 5.1.1.1.1. The pouch material shall not impart any undesirable odor or flavor to the product and shall be FDA approved for food use.

5.1.1.2.2 Pouch construction. The tray-shaped body and the tray-shaped cover shall be formed by drawing the flexible laminate material into an appropriately shaped cavity. One bar shall be placed into the tray-shaped body of the pouch. The filled pouch body shall be hermetically sealed in accordance with the vacuum requirements as specified in 5.1.1.1.3 and examined as specified in 4.5.5. Pouch closure shall be effected by heat sealing the tray-shaped cover to the tray-shaped body along the entire pouch perimeter. The outside dimensions of the sealed pouch shall be not more than 3-1/4 inches wide by 6-3/8 inches long. The closure seal width shall be a minimum 1/8 inch. Heat seals shall be free of foldover wrinkles and occluded matter. When tested as specified in 4.5.7, the pouches shall withstand an internal pressure of 14 psig for 30 seconds without rupture or seal separation greater than 1/16 inch or seal separation that reduces the effective closure seal width to less than 1/16 inch. A tear-notch shall be placed on one outside edge or two opposite outside edges of the pouch. Each tear-notch shall be located not more than 1-1/4 inches or less than 1/4 inch from an outside corner of the pouch. The tear notch shall be at least 1/32 inch deep. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. Alternatively, if the pouch has serrated edges, the serrations may be used as tear-notches provided that the serrations are located to effect easy opening in the machine direction of the pouch laminates; the serrations are sharp (no plastic tailing exists); and the serration depth and the minimum seal width at the serrations are in accordance with the above tear notch requirements. The sealed pouch shall not show any evidence of material degradation, aluminum stress cracking, delamination or foreign odor.

5.2 Packing. Packing shall be level C.

5.2.1 Level C packing. Not more than 40 pounds of style B cereal bars, preserved as specified in 5.1.1, shall be packed flat in layers in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, variety SW, grade 275, class domestic of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to grade II, class B or C of A-A-1898 or cushioning material not less than 1/16 inch thick conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual packages.

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5.3 Labeling and marking.

5.3.1 Pouches. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water insoluble, homogenous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low temperature. Dimensions of the label shall not be larger than the dimensions of the surface of the product where the label is applied on the package. Ink and adhesive used must be suitable for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 1/16 inch high. The label shall contain the following information:

Item Name 1/
Mixing Instructions 2/
Date 3/
Contractor's name

1/ Insert item name, as applicable:

SHREDDED WHEAT CEREAL BAR
CORN FLAKE CEREAL BAR
WHEAT FLAKES CEREAL BAR
MALTED WHEAT GRANULES CEREAL BAR
BRAN FLAKES CEREAL BAR
OAT CEREAL BISCUIT BAR

2/ Mixing instructions are as follows:

Eat as is, or if desired, rehydrate. To rehydrate, crumble bar, add small amount of water, stir.

3/ Each pouch shall have the date of pack noted by using a four digit code beginning with the final digit of the current year followed by the three digit Julian day code. For example, March 19, 1991 would read 1079. The Julian day code shall represent the date the product is unit packed into the pouch. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

Nutrition labeling shall be in accordance with the provisions of 21 CFR Part 101.9, Nutrition labeling of food.

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5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and style of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
f. Long Range Patrol or Food Packet, Survival, General Purpose component.

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Malted wheat cereal granules. It has been found that Grapenuts manufactured by Post Division of General Foods Corp., meets the requirements of 3.2.1d and performs satisfactorily in the type IV product.

6.3.2 Oat cereal biscuits. It has been found that Life cereal manufactured by Quaker Oats Co., meets the requirements of 3.2.1f and performs satisfactorily in the type VI product.

6.3.3 Lecithin, liquid. It has been found that Centrophase C Special Grade Lecithin manufactured by Central Soya Company, Inc., meets the requirements of 3.2.6 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Food Packet, Long Range Patrol, (Improved)
Food Packet, Survival, General Purpose
Pouched food
Shelf stable

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8920-0550)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44389A	1993 September 24

3. DOCUMENT TITLE

CEREAL BARS, (OPERATIONAL RATION COMPONENT)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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